

1 Range of Coverage Analysis and Material Availability

Overview

Menu	Material management → Inventory management → Estimation of range of coverage
Transaction code	roc
Function authorization	roc

Usage

This function analyzes the range of coverage of the material transferred between two consecutive production levels (preceding level and subsequent level).

For generating the list, materials are considered which are

- currently produced in the preceding level
- used as components in the subsequent level

For each material, the number of supplying as well as retrieving machines is now indicated. By means of option buttons, only the supplying or only the consuming machines can be considered.

The following criteria are relevant for the range of coverage:

- The consuming OPs and their target cycle result in a demand per unit of time.
- The supplying OPs and their target cycle result in a demand increase per unit of time.

The current stock of material buffers to be analyzed is considered as initial stock. In this regard, batches currently logged on to consuming machines are also considered.

Integration

For the purpose of analyzing material coverage ranges and/or material availability, so-called production levels are defined. In this process, various machines as well as material buffers from where the machines of a specific production level may retrieve material are allocated to a specific production level.

In addition, the supply relationship between production levels is shaped. A production level may have several subsequent and/or preceding levels. In this supply relationship, a minimum coverage range is defined.

The range of coverage computation is requested for a production level considered as consuming in this context. With regard to the analysis, all supply levels/production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, the latter are also included for the purpose of a holistic analysis.

Prerequisite

The production levels and supply relationships must exist in the system.

Selection criteria

Production levels

The range of coverage computation is requested for a production level considered as retrieving in this context. With regard to the analysis, all supply levels/production levels assigned to this level as preceding levels are then considered. If these, in turn, are supply levels to other levels, these other, retrieving production levels are also included for the purpose of a holistic analysis.

Incl. materials (currently) only supplied/produced

This option allows for extending the analysis to also include materials which are currently only supplied and/or only produced.

Display

The range of coverage is shown in hours in the screen. For a cyclical display, the data are updated every 180 seconds. The following data are displayed:

Material

The material corresponds to the materials in the component list in producing operations and to the article ID in supplying operations, respectively.

Material designation

Material designation

Number of retrieving machines (total)

Machines within the supply relationships of the current production level and with currently running operations using this material.

In production

Machines within the supply relationships of the current production level and with currently running operations using this material and operating in the *Production* status.

Number of supplying machines (total)

Machines within the supply relationships of the current production level and with currently running operations producing this material.

In production

Machines within the supply relationships of the current production level and with currently running operations producing this material and operating in the *Production* status.

Range of coverage

The range of coverage in [HH:MM:SS] is computed as follows:

In the first step for each operation and material component, the quantity per hour is computed. The basis for this is the target cycle entered for the operation (= speed at which the article is produced), as well as the input quantity of the included material components.

Since an article may be processed at different target cycle times in different orders/operations, the individual values are added up.

The basis for computing the range of coverage of a material is the remaining quantity entered in the batch stock (with material number). The batches considered are those in the material buffers of the immediately preceding level (acc. to configuration). Only batches in the F (free) and L (running) status are considered.

The range of coverage [in hours] is determined from the ratio of the values described above:

- Remaining quantity acc. to batch stock
- Required quantity per hour.

For materials which are indicated in the list (because they are currently produced on a supplying machine), but which are not retrieved (because no retrieving order/operation is active), the range of coverage is not computed.

For materials which are more rapidly produced than consumed, the range of coverage is also infinite.

In these cases, the "infinite" symbol is displayed in the "Trend" column.

Trend

In addition, the trend of the range of coverage (change in comparison to previous state) is indicated in front of each bar by an appropriate symbol.

- increasing 
- decreasing 
- unchanged 
- infinite ∞

On the first data request, the "unchanged" symbol is displayed.



In the range of coverage analysis, the  function can be used to switch to the "Stock overview" in order to display the stock overview of the previously selected material there.



Each of the materials (batches) and material components must be defined in the same quantity unit; there is no internal quantity conversion.